

RK vs NN proton reconstruction on breakup g4output

Head-to-head evaluation

2026-06-18

Abstract

This report evaluates proton reconstruction quality on the d+Sn breakup sample (ypol + zpol, 176 cells) using two backends: RK (Runge-Kutta least-squares fitter, three-point-free mode, run on spana03) and NN (6-128-128-64 MLP, formal_B115T3deg/20260420_184345 model, run locally). Both reconstructions derive from the same Geant4 g4output, enabling per-event entry-index alignment. Three error sets are reported for protons (RK-vs-truth, NN-vs-truth, RK-vs-NN), plus truth-vs-reco residuals for neutrons (NEBULA β -method, identical for both backends). Sample: 3301419 truth-proton events total.

1 Data & method

RK reco: spana03 data/reconstruction/dbreak_elastic/, 176 files, produced 2026-05-09 via `run_dbreak_elastic_pipeline.py stage_reco` with `--backend rk --rk-fit-mode three-point-free --max-iterations 40`.

NN reco: local data/reconstruction/breakup_nn_20260503/, 176 files, produced 2026-05-03 via `01_run_nn_reco.sh` with the clean production model `formal_B115T3deg/20260420_184345`.

g4output source: identical for both backends (verified by file size 285252607 bytes, timestamp 2026-05-01 20:16:46 +0900). Per-event entry-index alignment is valid 1:1.

Frame convention: target frame (beam-as-Z) throughout. Truth is already target-frame; both backends' reco is rotated to target frame at write time via `RotateRecoResultToTargetFrame` (`apps/run_reconstruction/main.cc:629-630`). No additional rotation in this analysis.

2 Proton reconstruction — RK vs NN vs truth

2.1 Global residuals

Pol	Backend	Δp_x	Δp_y
ypol	RK	MAE=11.4, RMSE=39.7, Bias=-6.8, P95=30.7, N=2150443	MAE=1.6, RMSE=3.5, Bias=-0.1, P95=4.1, N=2150443
ypol	NN	MAE=6.1, RMSE=13.3, Bias=-1.8, P95=14.4, N=2150443	MAE=1.3, RMSE=2.4, Bias=+0.5, P95=3.1, N=2150443
zpol	RK	MAE=12.0, RMSE=46.8, Bias=-7.4, P95=35.3, N=101662	MAE=1.7, RMSE=4.8, Bias=-0.2, P95=3.1, N=101662
zpol	NN	MAE=7.1, RMSE=13.7, Bias=-2.6, P95=22.2, N=101662	MAE=1.4, RMSE=3.0, Bias=+0.6, P95=2.1, N=101662

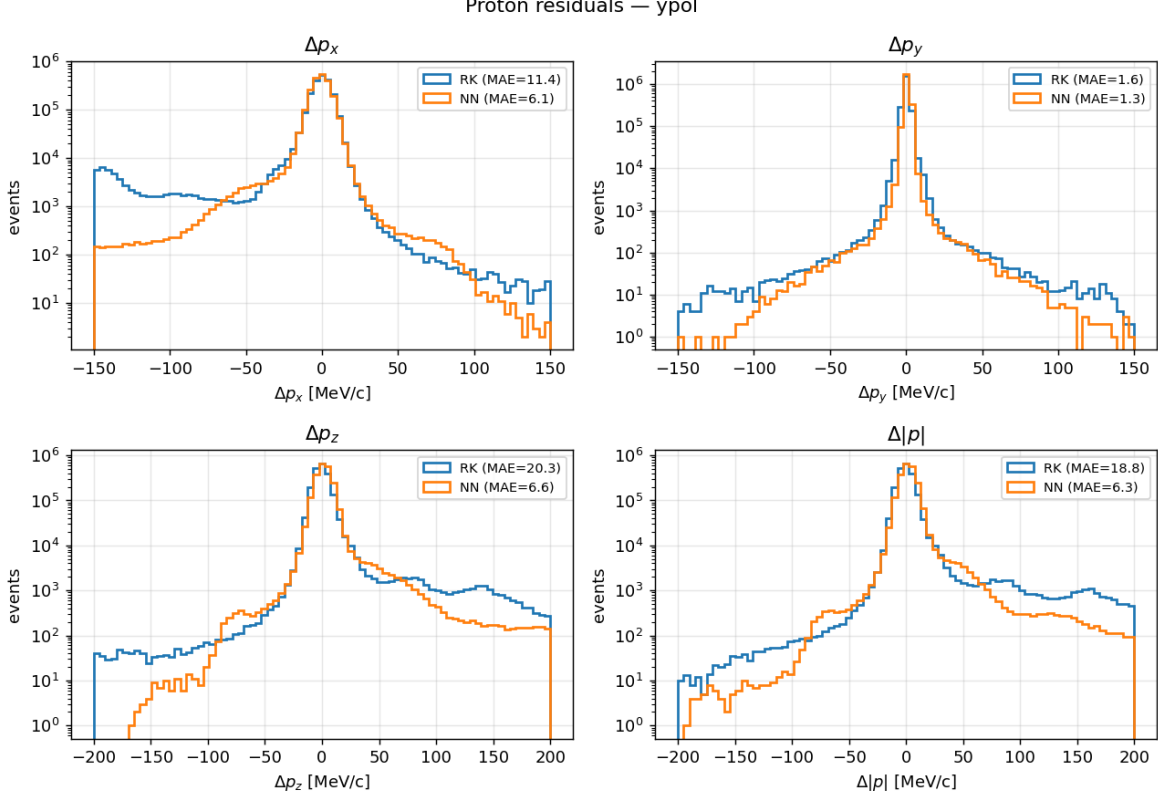


Figure 1: Proton residual distributions, ypol.

2.2 Per-event agreement (RK vs NN)

For each event, compare $|p_{RK} - p_{truth}|$ vs $|p_{NN} - p_{truth}|$. Points below the $y = x$ line are events where RK is closer to truth; above means NN is closer.

2.3 Per-cell breakdown

2.4 RK uncertainty calibration (pull)

Pull = $(|p|_{RK} - |p|_{truth})/\sigma_p$. Should be $\sim \mathcal{N}(0, 1)$ if Fisher uncertainties are calibrated.

3 Neutron reconstruction — truth vs reco

Neutron momentum is reconstructed by the NEBULA β -method in both RK and NN backends (the backend choice only affects proton reconstruction). Residuals below are therefore single-backend (NN file used).

4 Conclusion

(To be filled in after inspecting the figures.)

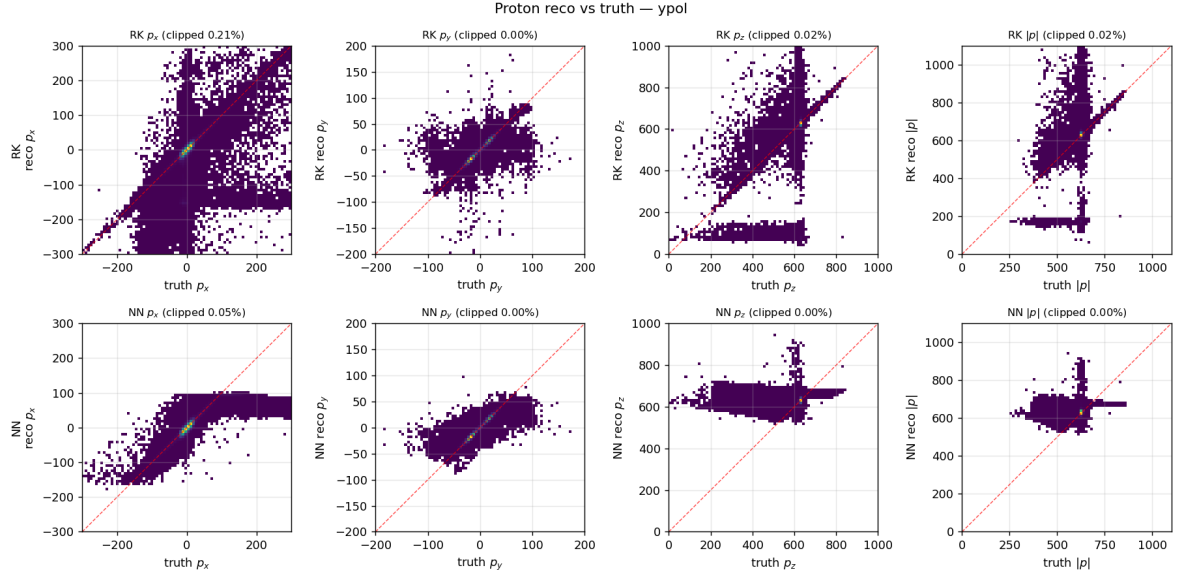


Figure 2: Proton reco vs truth 2D scatter, ypol.

A Per-cell numeric table

(See `summary.json` for full per-cell numeric breakdown.)

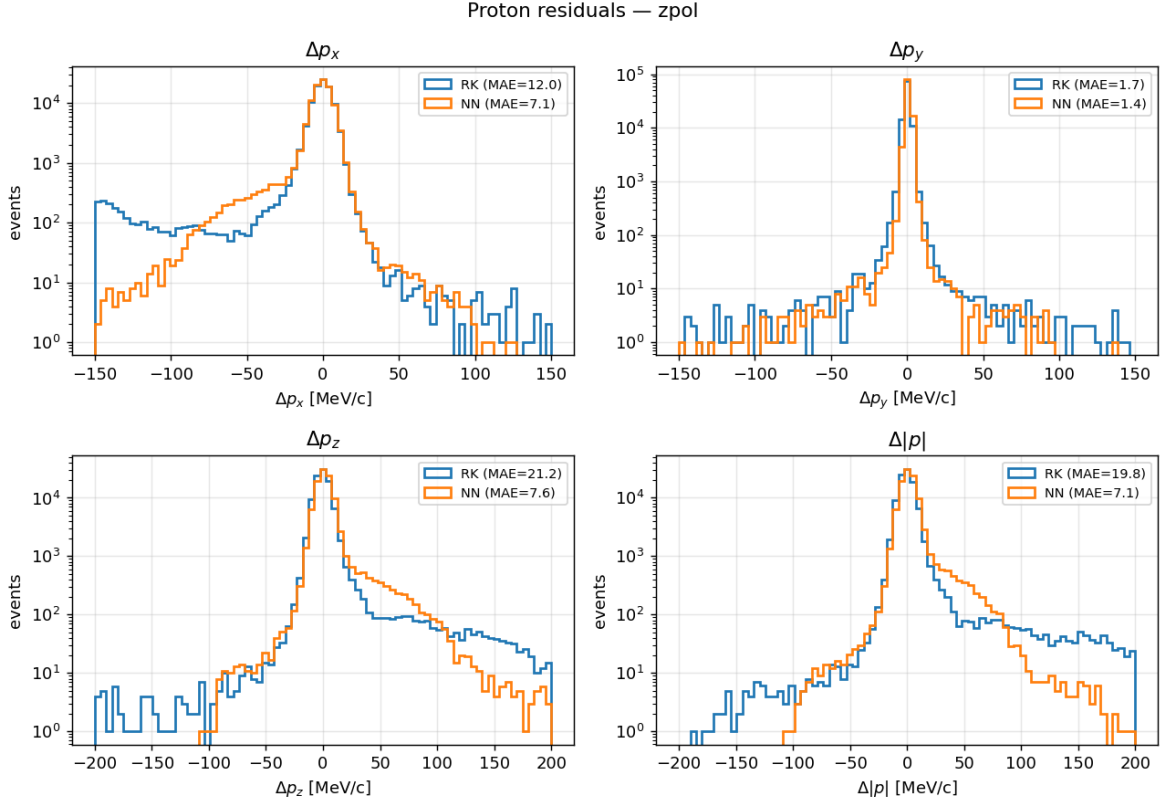


Figure 3: Proton residual distributions, zpol.

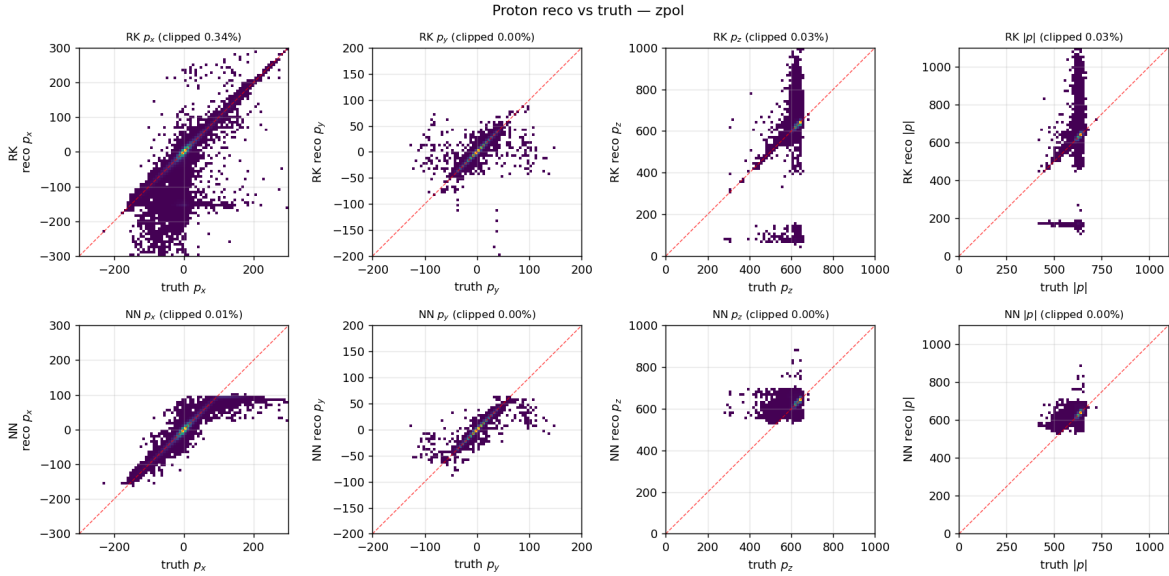


Figure 4: Proton reco vs truth 2D scatter, zpol.

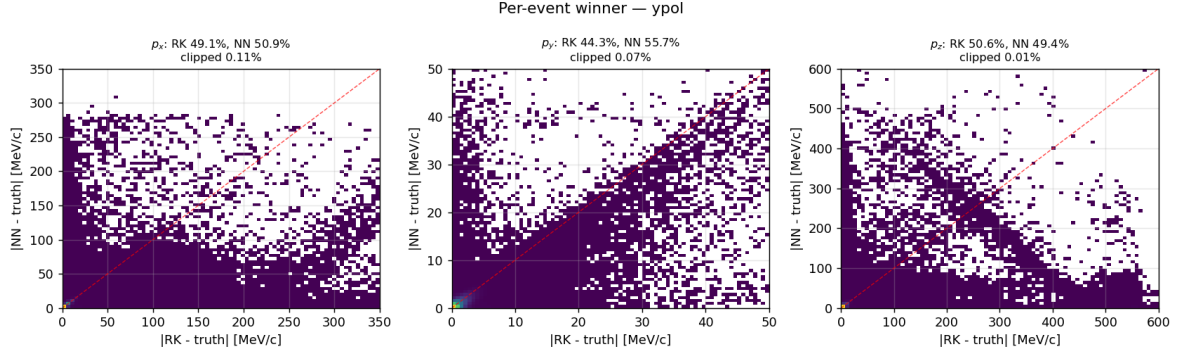


Figure 5: Per-event winner, ypol.

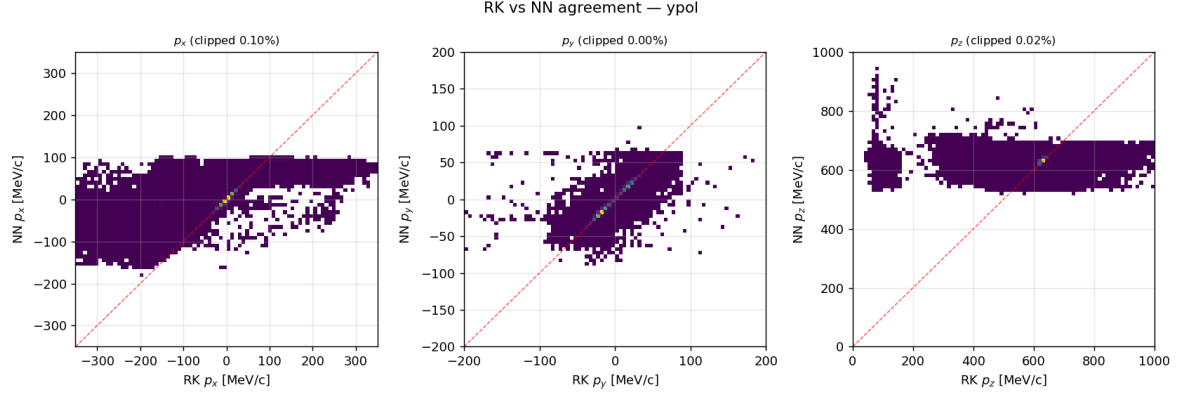


Figure 6: RK vs NN direct agreement, ypol.

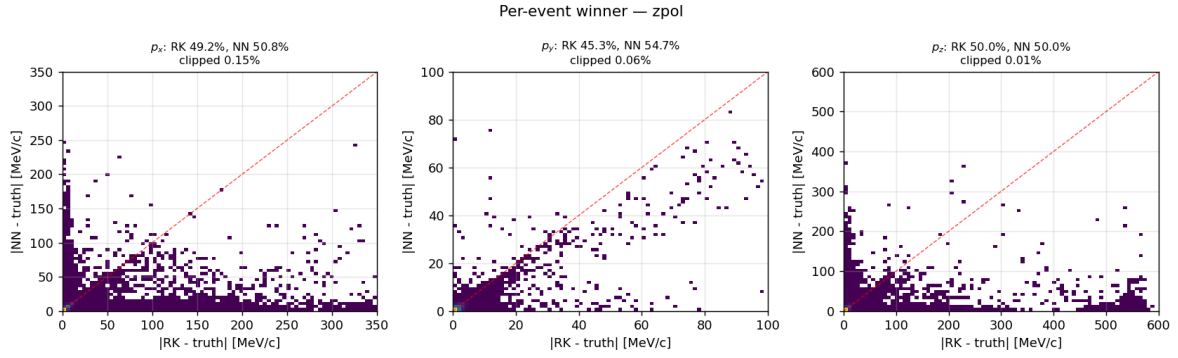


Figure 7: Per-event winner, zpol.

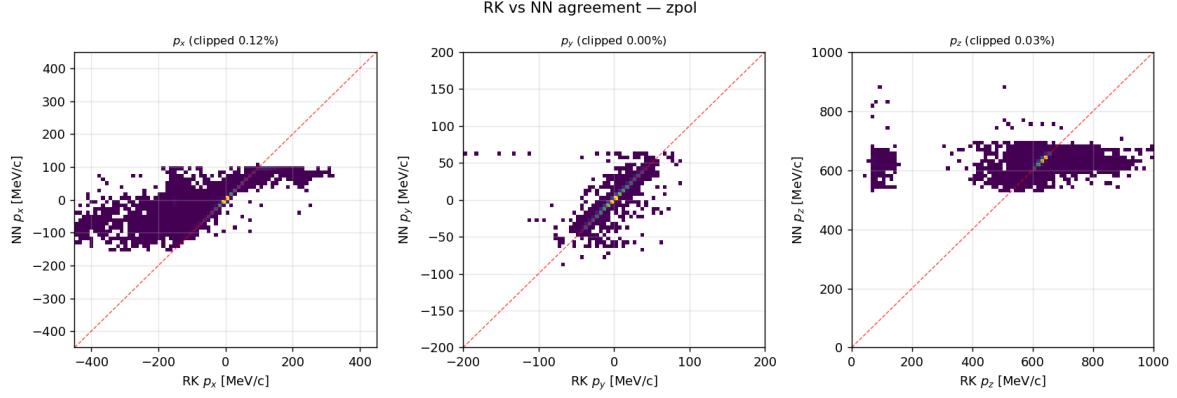


Figure 8: RK vs NN direct agreement, zpol.

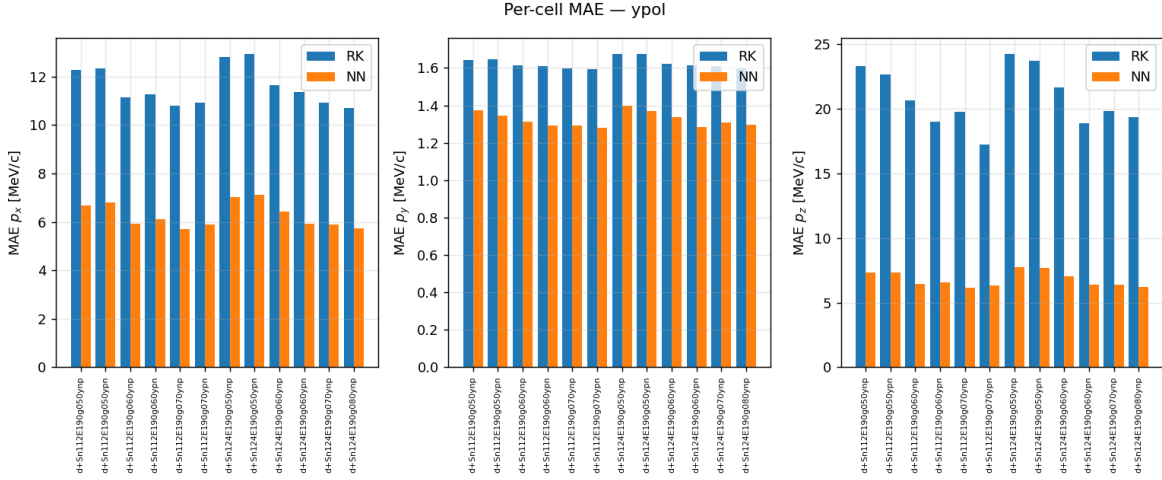


Figure 9: Per-cell MAE, ypol.

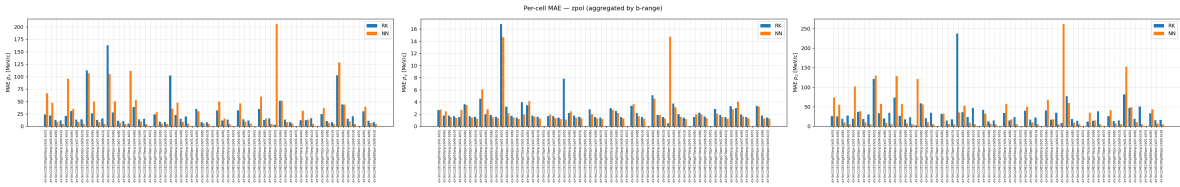


Figure 10: Per-cell MAE, zpol (aggregated by b-range).

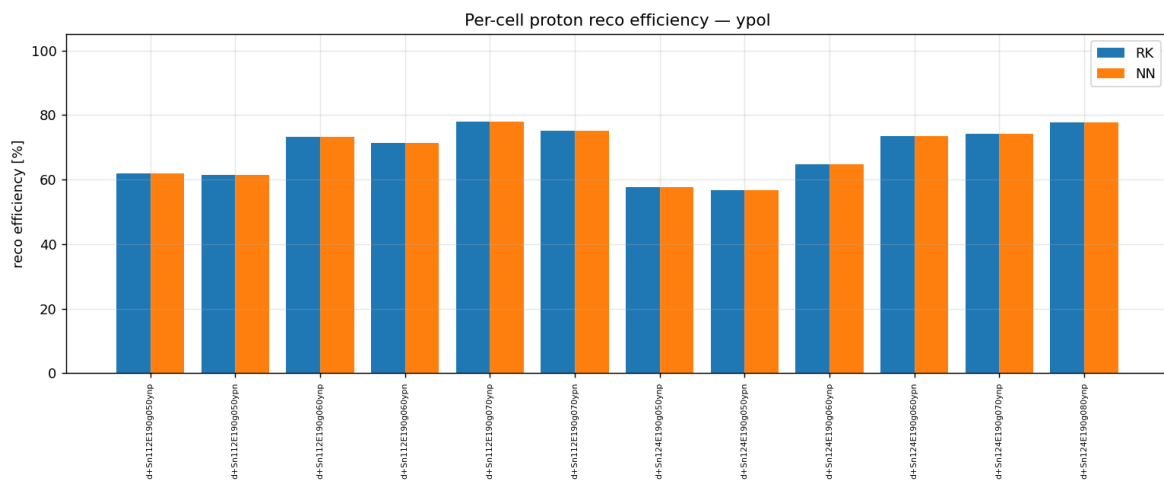


Figure 11: Per-cell proton reco efficiency, ypol.

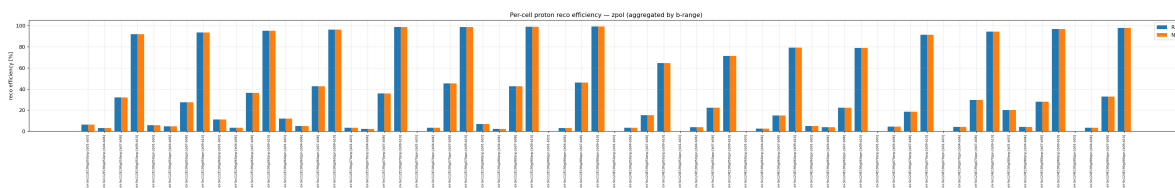


Figure 12: Per-cell proton reco efficiency, zpol (aggregated by b-range).

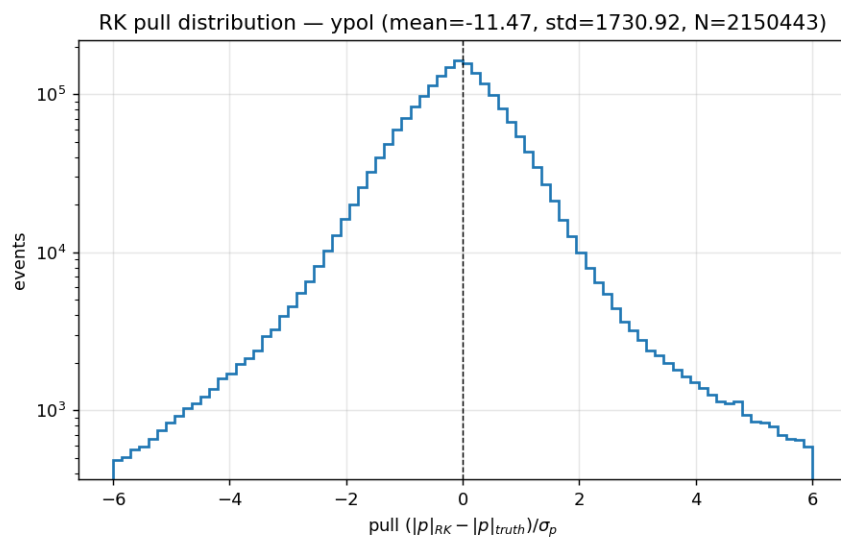


Figure 13: RK pull distribution, ypol.

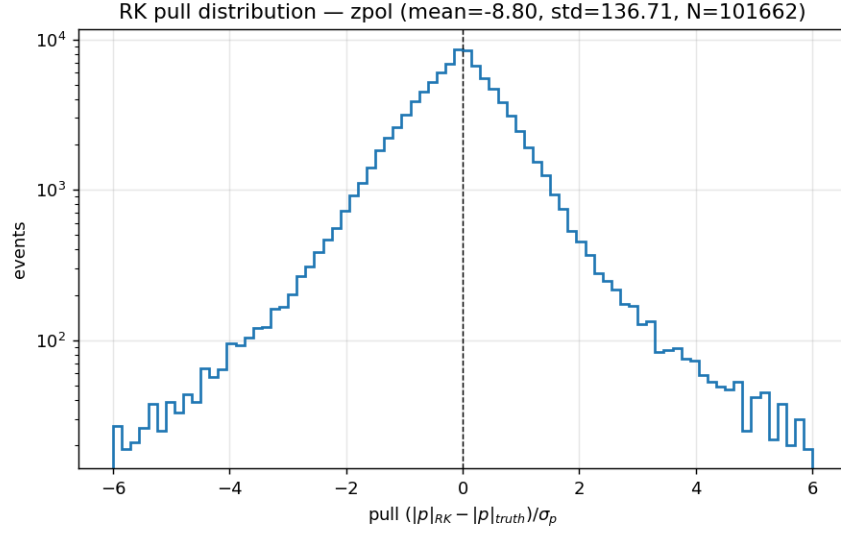


Figure 14: RK pull distribution, zpol.

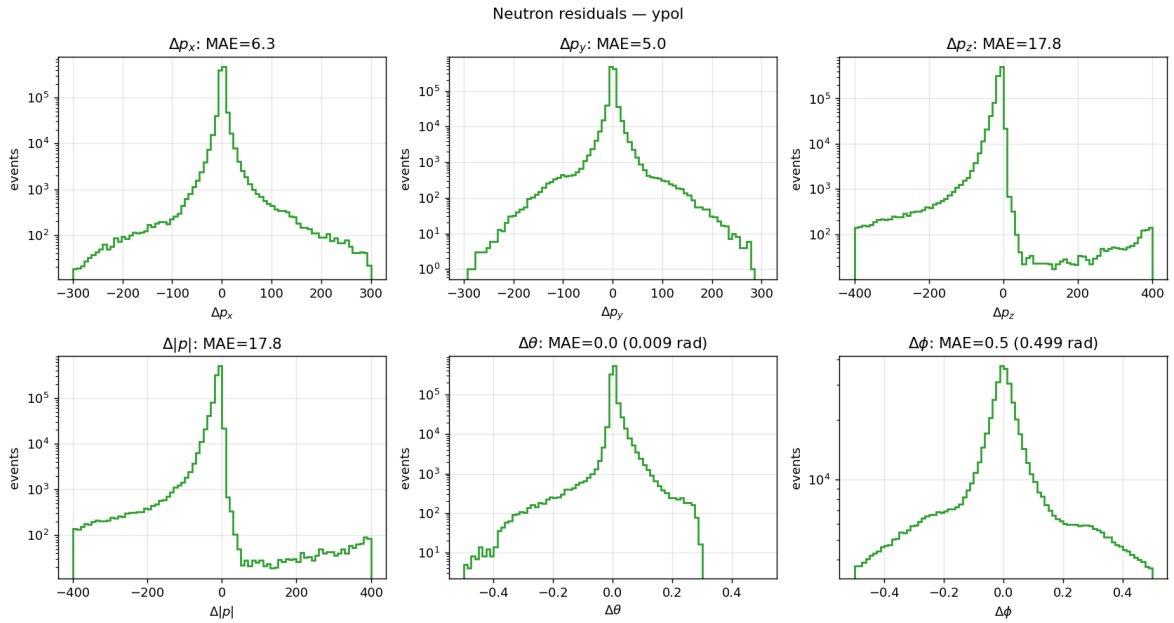


Figure 15: Neutron residual distributions, ypol.

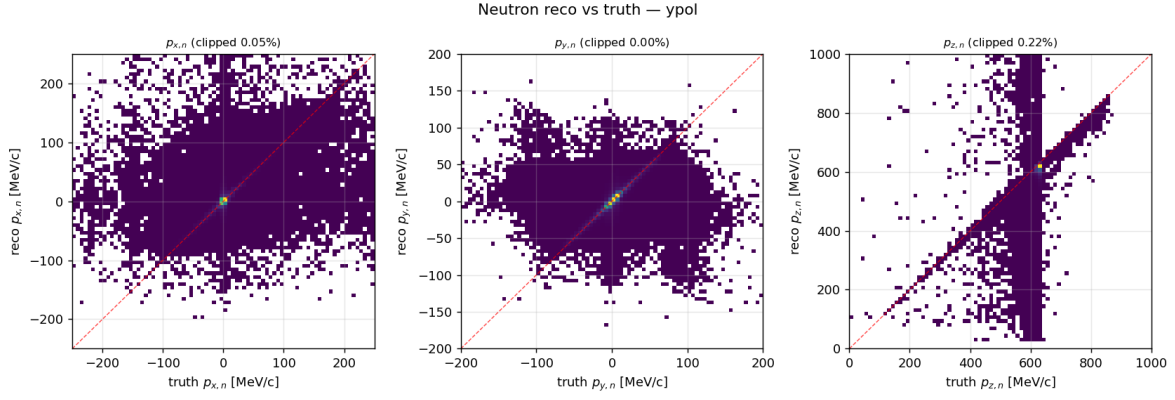


Figure 16: Neutron reco vs truth 2D scatter, ypol.

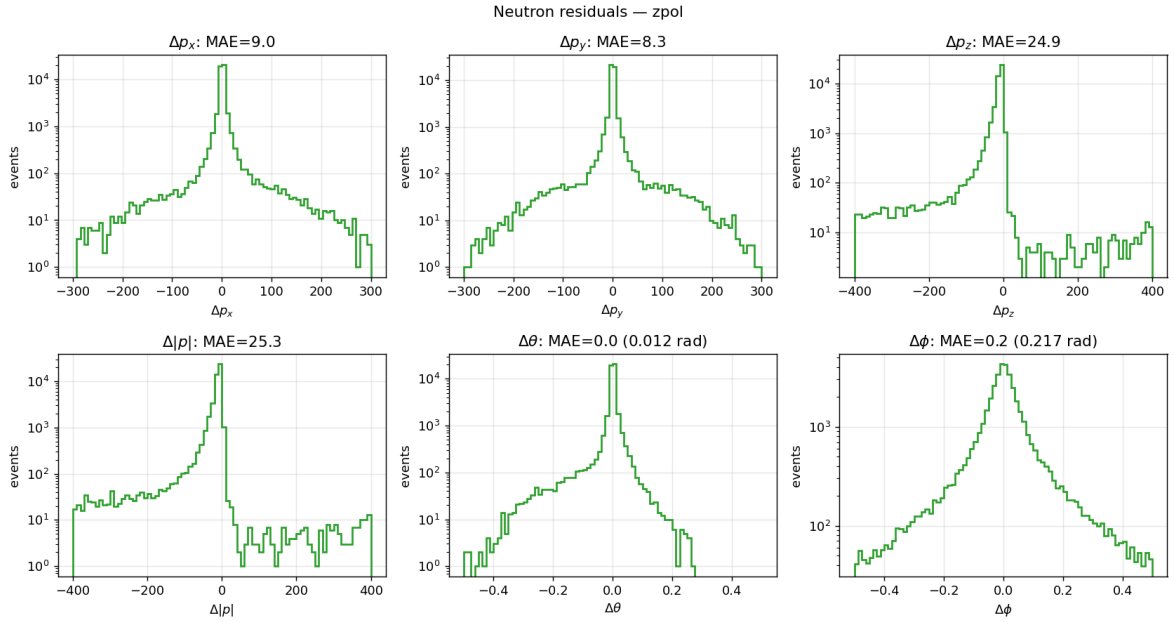


Figure 17: Neutron residual distributions, zpol.

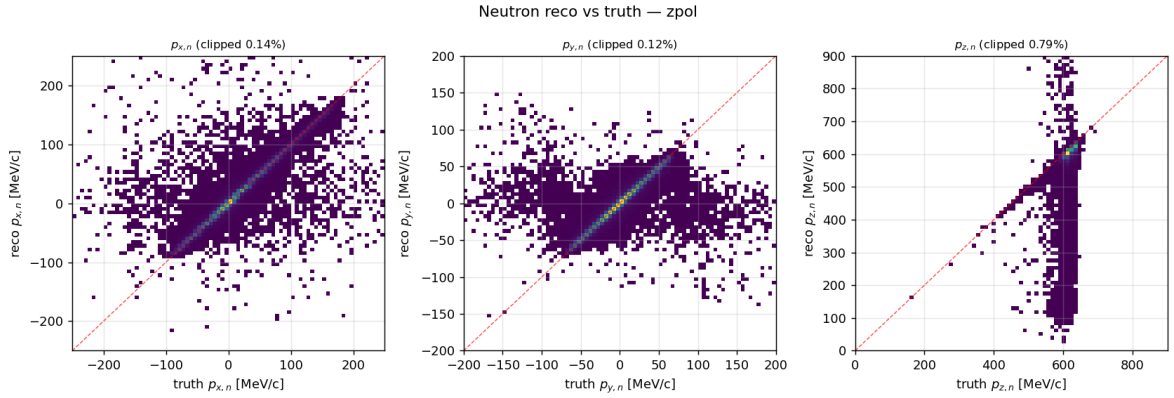


Figure 18: Neutron reco vs truth 2D scatter, zpol.